

File & Voice Security Documentation

Enhancing Security in Messaging Applications

Spinnaker Analytics

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1. Introduction

1.1 Purpose

- This document defines the security controls implemented to protect file transfers and voice communications within the messaging application.
- The objective is to ensure confidentiality, integrity, authenticity, and secure storage of user-generated content while protecting voice communications against interception, tampering, and unauthorized access.

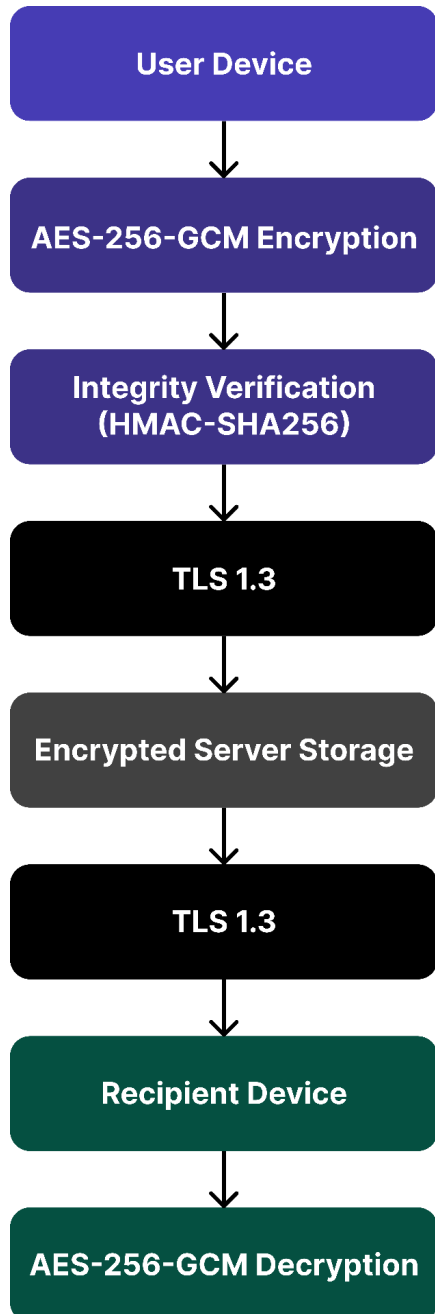
1.2 Objectives

- Protect files before transmission
- Secure file storage on client and server systems
- Ensure encrypted file transfers
- Secure real-time voice communications
- Prevent unauthorized access to stored content
- Maintain message and call confidentiality
- Validate security controls through testing

2. File Security Architecture

Files are protected using a layered security model.

Figure 15 – File Security Architecture



2.1 Security Layers

Layer	Protection
Application Layer	AES-256-GCM
Integrity Layer	HMAC-SHA256
Transport Layer	TLS 1.3
Storage Layer	SQLCipher
Device Layer	Full Disk Encryption

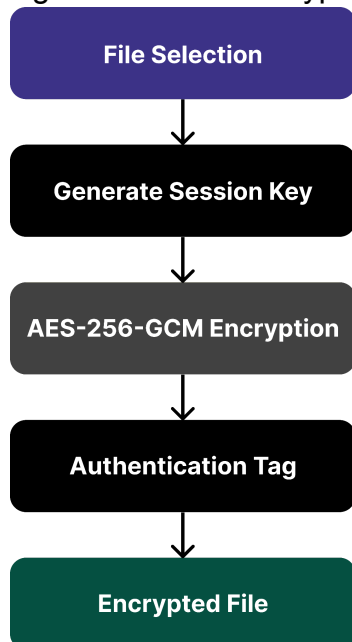
3. Secure File Encryption

3.1 Encryption Process

- Before transmission:
 - File selected
 - Unique encryption key generated
 - AES-256-GCM encryption applied
 - Integrity tag generated
 - Ciphertext transmitted

3.2 File Encryption

Figure 16 – File Encryption Workflow



3.3 Encryption Parameters

Parameter	Value
Algorithm	AES-256-GCM
Key Length	256-bit
Nonce Length	96-bit
Authentication Tag	128-bit
Key Generation	CSPRNG

3.4 Security Benefits

- Confidentiality
- Integrity
- Tamper detection
- Replay protection

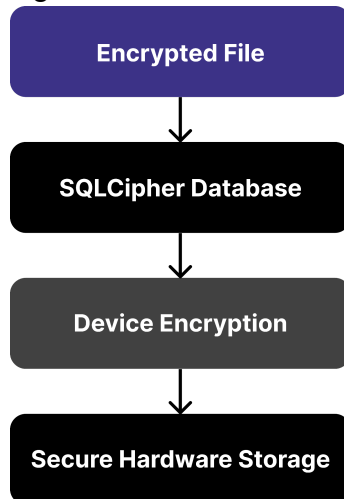
4. Secure File Storage

4.1 Client-Side Storage

- Encrypted files stored locally using:
 - **Android**
 - Android Keystore
 - Full Disk Encryption
 - SQLCipher
 - **iOS**
 - Secure Enclave
 - Data Protection API
 - Keychain Services

4.2 Secure Storage

Figure 17 – Secure Storage Architecture



4.3 Server-Side Storage

- Server storage remains encrypted.
 - Controls
 - AES-256 encrypted storage
 - Encrypted backups
 - Metadata separation
 - Access logging

Storage Location	Protection
Android Device	Keystore + FDE
iOS Device	Secure Enclave
SQLCipher Database	AES-256
Server Storage	AES-256
Backups	Encrypted

4.4 Data Classification

Data Type	Protection
Files	AES-256
Metadata	Restricted Access
Logs	Retention Policy
Backups	Encrypted

5. File Transfer Security Controls

5.1 Transport Security

- All transfers occur through:
 - TLS 1.3

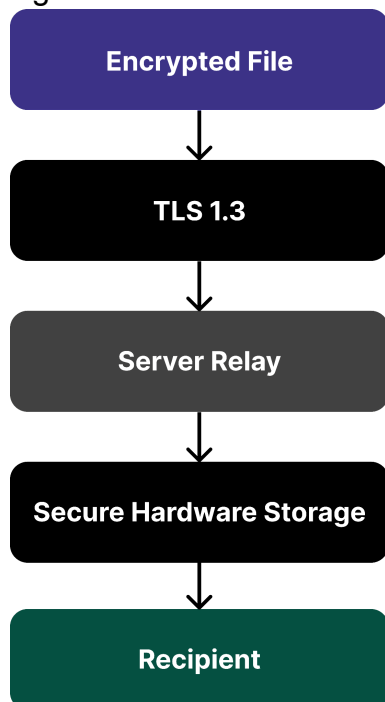
5.2 Cipher Suites

- Cipher Suite
 - TLS_AES_256_GCM_SHA384
 - TLS_CHACHA20_POLY1305_SHA256

5.3 Security Controls

- **Certificate Validation**
 - Trusted CA verification
 - Certificate chain validation
- **Perfect Forward Secrecy**
 - Implemented using:
 - **ECDHE**

Figure 18 – Secure File Transfer Process



6. Voice Communication Security

- Voice communication security is implemented using secure real-time communication protocols.

6.1 Security Objectives

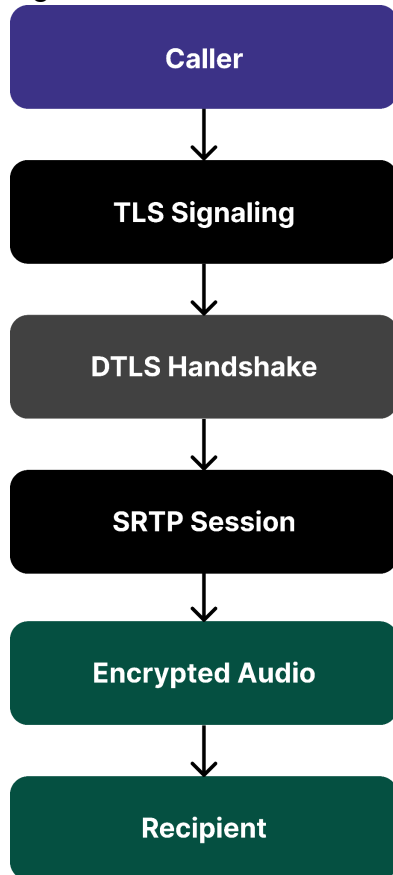
- Confidentiality
- Integrity
- Authentication
- Forward Secrecy
- Real-Time Performance

6.2 Voice Security Components

Component	Technology
Signaling	TLS 1.3
Key Exchange	DTLS
Voice Encryption	SRTP
Key Agreement	ECDHE
Integrity Protection	SRTP Authentication

7. Secure Voice Call Architecture

Figure 19 – Secure Voice Call Architecture



7.1 Call Establishment Process

- **Step 1**
 - Call Initiation
- **Step 2**
 - TLS Signaling
- **Step 3**
 - DTLS Handshake
- **Step 4**
 - ECDHE Key Exchange
- **Step 5**
 - SRTP Session Creation
- **Step 6**
 - Encrypted Audio Streaming

8. Voice Call Protection Controls

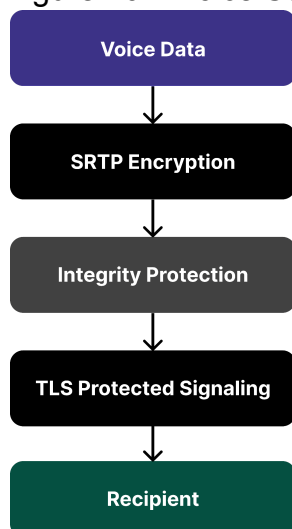
8.1 Encryption Parameters

Parameter	Value
Protocol	SRTP
Cipher	AES-128-GCM
Key Exchange	DTLS
Forward Secrecy	ECDHE
Authentication	SRTP Integrity Checks

8.2 Security Controls

- Protection Against Eavesdropping
 - SRTP Encryption
 - TLS Signaling
- Protection Against Replay Attacks
 - Sequence Numbers
 - Session Validation
- Protection Against Session Hijacking
 - DTLS Authentication
 - Ephemeral Session Keys

Figure 20 – Voice Security Layers



9. Security Validation & Testing

9.1 Testing Environment

Tool	Purpose
Wireshark	Traffic Analysis
Burp Suite	TLS Verification
MobSF	Mobile Security Assessment
OpenSSL	Certificate Validation

9.2 Validation Objectives

- **File Security**
 - Verify:
 - File encrypted before upload
 - Encrypted storage active
 - No plaintext transfer
- **Voice Security**
 - Verify:
 - SRTP traffic present
 - No plaintext audio
 - TLS signaling active

9.3 Test Cases

Test ID	Test Scenario	Expected Result
FILE-01	Upload file	Encrypted file
FILE-02	Download file	Successful decryption
FILE-03	Inspect traffic	Ciphertext only
VOICE-01	Voice call	SRTP traffic
VOICE-02	TLS inspection	TLS 1.3 verified
VOICE-03	Replay simulation	Rejected

9.4 Expected Evidence

- Wireshark packet captures
- TLS handshake screenshots
- MobSF assessment reports
- Encrypted storage screenshots

10. Security Considerations

10.1 Threats & Mitigations

Threat	Mitigation
File Interception	AES-256-GCM
Data Tampering	HMAC-SHA256
MITM Attacks	TLS 1.3
Replay Attacks	Nonces
Voice Eavesdropping	SRTP
Session Hijacking	DTLS

10.2 Standards Alignment

Standard	Purpose
RFC 3711	SRTP
RFC 8446	TLS 1.3
NIST SP 800-57	Key Management
OWASP MASVS	Mobile Security
OWASP MSTG	Mobile Testing

11. Conclusion

- This File & Voice Security Documentation defines the controls used to secure file transfers and voice communications throughout their lifecycle.
 - The solution combines:
 - AES-256-GCM file encryption
 - SQLCipher secure storage
 - TLS 1.3 transport protection
 - DTLS key exchange
 - SRTP voice encryption
 - Perfect Forward Secrecy
 - Comprehensive validation testing
- These controls ensure confidentiality, integrity, and availability of user communications while protecting against modern cyber threats such as interception, tampering, replay attacks, and unauthorized access.